[Diggavi 2004]. 802.11ac and 802.11 ax base stations may transmit to multiple stations simultaneously, and use "smart" antennas to adaptively beamform to target transmissions in the direction of a receiver. This decreases interference and increases the distance reached at a given data rate. The data rates shown in Table 7.1 are for an idealized environment, for example, a receiver close to the base station, with no interference—a scenario that we're unlikely to experience in practice! So as the saying goes, YMMV: Your Mileage (or in this case your wireless data rate) May Vary.

7.3.1 The 802.11 Wireless LAN Architecture

Figure 7.7 illustrates the principal components of the 802.11 wireless LAN architecture. The fundamental building block of the 802.11 architecture is the **basic service set (BSS)**. A BSS contains one or more wireless stations and a central **base station**, known as an **access point (AP)** in 802.11 parlance. Figure 7.7 shows the AP in each of two BSSs connecting to an interconnection device (such as a switch or router), which in turn leads to the Internet. In a typical home network, there is one AP and one router (typically integrated together as one unit) that connects the BSS to the Internet.

As with Ethernet devices, each 802.11 wireless station has a 6-byte MAC address that is stored in the firmware of the station's adapter (that is, 802.11 network interface card). Each AP also has a MAC address for its wireless interface. As with Ethernet, these MAC addresses are administered by IEEE and are (in theory) globally unique.

As noted in Section 7.1, wireless LANs that deploy APs are often referred to as **infrastructure wireless LANs**, with the "infrastructure" being the APs along with the

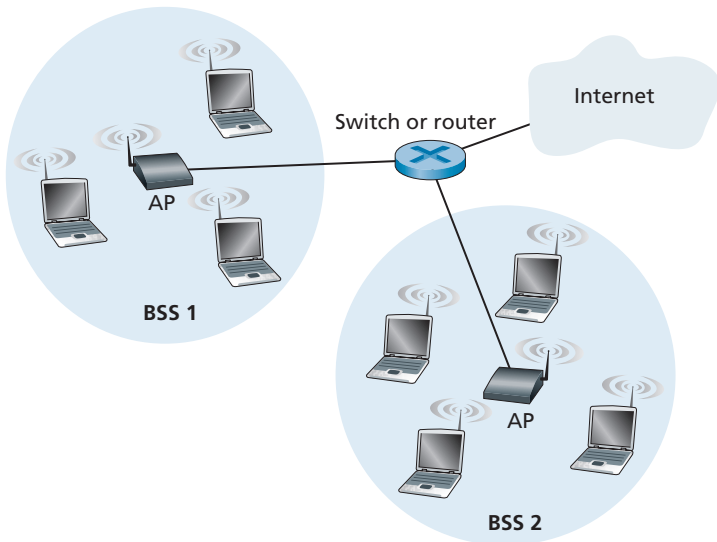

Figure 7.7 ♦ IEEE 802.11 LAN architecture